

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	43.15912	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	85.13068	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1486.83892	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.152817	0.152817	242.93476	0.53038	15.6575

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 43.15912.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.

- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 85.13068.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1486.83892.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
 - X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.197631.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.152817.
- Held-out runtime ms: 242.93476.
- Robustness dispersion across held-out structure families: 0.003803.
- Worst held-out instances:
 - X-n190-k8: feasible=True, penalized gap=0.270141, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.176371, errors=[]
 - X-n247-k50: feasible=True, penalized gap=0.120137, errors=[]

- X-n223-k34: feasible=True, penalized gap=0.113337, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.0841, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver-Evolution Suite

Metric	NN Constructive	Clarke-Wright Savings	Regret Insertion LS	Evolved Solver
Feasibility Rate	1.0 (Heldout & Train)	1.0 (Heldout & Train)	1.0 (Heldout & Train)	1.0 (Heldout & Train)
Objective Gap (Heldout)	0.2734	**0.0738 (best)**	0.4664	0.1528
Runtime (Heldout ms)	43.16	85.13	1486.84	242.93
Robustness Dispersion	0.00278	0.00779	0.08690	0.00380
Mean Complexity	0.0	0.0	0.0	15.66
Code Novelty	0.0	0.0	0.0	0.53

Conservative Analysis

- ****Feasibility:**** All methods achieved perfect feasibility (1.0) on held-out instances.
- ****Objective Gap:**** The Clarke-Wright baseline achieved the lowest objective gap by a clear margin, outperforming the evolved solver (0.0738 vs. 0.1528).
- ****Runtime:**** Nearest Neighbor was fastest (~43 ms), evolved solver runtime (~243 ms) is moderate, much faster than regret insertion (~1487 ms).
- ****Robustness:**** The evolved solver showed low robustness dispersion (0.0038), close to nearest neighbor and better than Clarke-Wright and regret insertion.
- ****Complexity & Novelty:**** Only the evolved solver introduced substantial code novelty and complexity.

Conclusion

The evolved solver maintained perfect feasibility and showed improved robustness compared to most baselines (except nearest neighbor). However, it did ****not clearly beat the best fixed baseline**** (Clarke-Wright Savings) in objective gap or runtime, despite introducing novel and more complex strategies. It represents a promising intermediate solution but does not definitively surpass classical heuristics on held-out structural families.